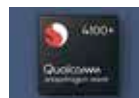




Writing with AuraRing

Watch this video to see how AuraRing allows one to write by merely using finger gestures. <http://dgit.in/sep20-91>



New Snapdragon Wear platform

Learn more about the Snapdragon Wear 4100 and 4100+ SoCs that will power future wearables. <http://dgit.in/sep20-92>

Next Gen Wearables

We are going to become cyborgs

Aditya Madanapalle | aditya@dgit.in

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hen it comes to wearables, we have a few solid devices that are not very convenient and not

good at what they do. There is considerable research into flexible, stretchable electronics, as well as new e-textiles that can greatly improve the range of the things that we use everyday that can become smart.

There is also research into translating the range of human movements into virtual space, for the next generation of interfaces, as well as potential applications in XR spaces. Many of these technologies also have potential medical applications.

TRACKING GESTURES

One of the untapped areas of human computer interaction are gestures. Human hands and fingers can be as articulate as our tongues, and so far we have only seen this on flat 2D surfaces, or to a limited extent in three dimensions. Researchers from the University of Washington have developed the AuraRing, which combines with a wrist-based tracker to feed in finger-based gestural inputs to a computer. The ring can track the 3D orientation of a finger, and continuously track hand movements. The ring is made up of a copper coil



Electricity harvesting fabric can power up other wearables

around a 3D printed ring. A current running through the ring generates a magnetic field, which is tracked by sensors on the wristband. The combination allows the band to continuously monitor the position and orientation of the ring, and by extension, the finger. The smart part is that the researchers got away with the demanding power requirements of wireless transmissions, and instead designed the system in such a way that the ring does not need to communicate with the wristband at all.



The AuraRing

Instead, the wristband only tracks the magnetic field generated by the ring. The setup works even when the ring is out of sight, such as in a crowded bus during commute, the ring can be used to trace out short messages in the air without having to retrieve the smartphone from the pockets. The ring can detect taps, flicks and pinches in thin air, and can even differentiate between a small pinch and a big pinch. For example once a response is traced out, a flick can be used to send the message. The ring has biomedical applications as well, as it is sensitive enough to detect the subtle tremors that signal the onset of Parkinson's disease, or monitor the rehabilitation process of those who have suffered from a stroke. Facebook and Google also participated in the research.

A team of researchers from the University of Wisconsin-Madison and Cornell University have devel-